H1N1 VACCINE PREDICTION

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Overview

Objectives

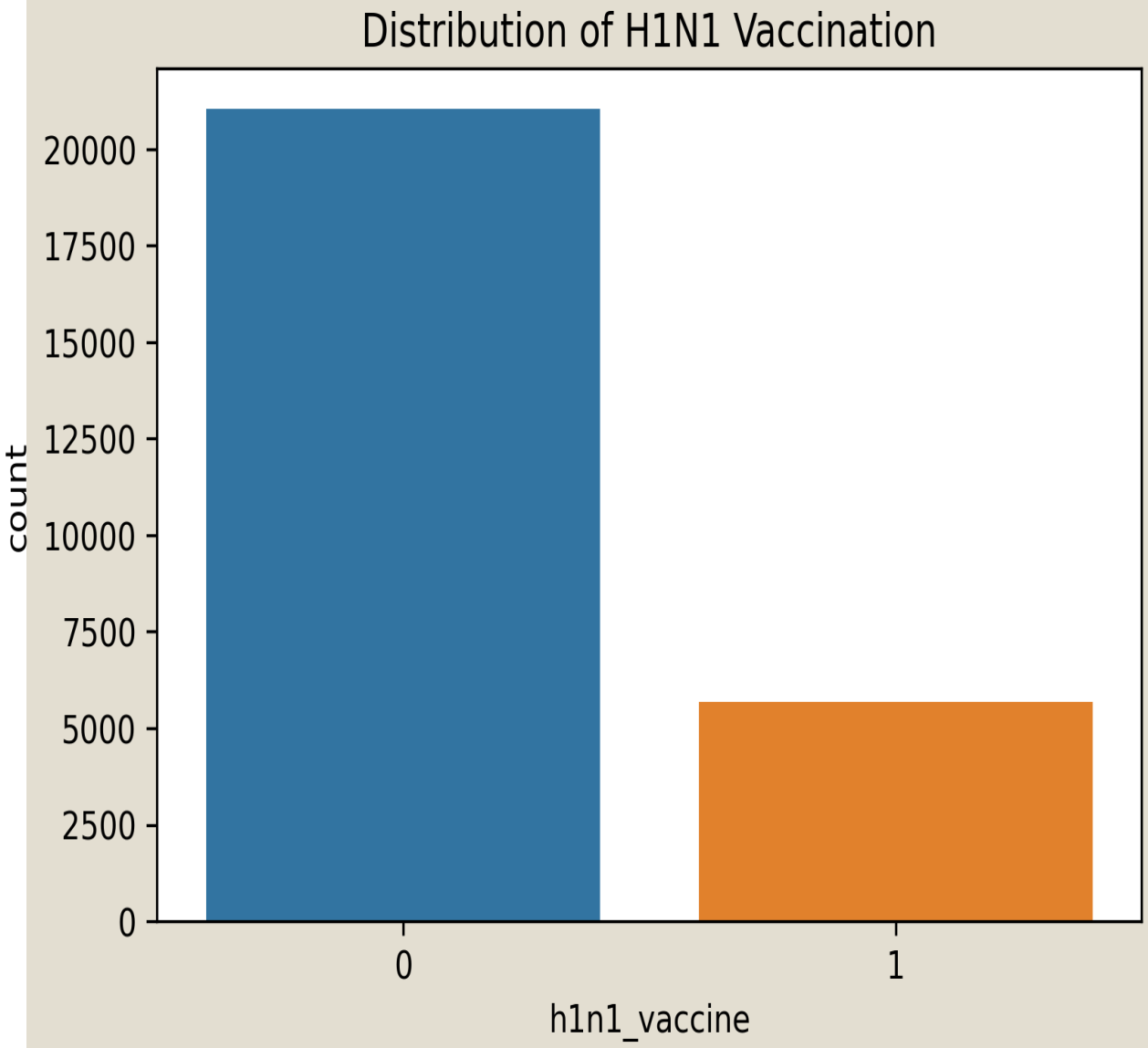
- Predict vaccination behavior.
- Identify its key influencing factors.

Business Problem

- Public health organizations need better ways to identify individuals who are likely or unlikely to get vaccinated.
- Public health organizations need better ways to target individuals who are likely or unlikely to get vaccinated.

Data Overview

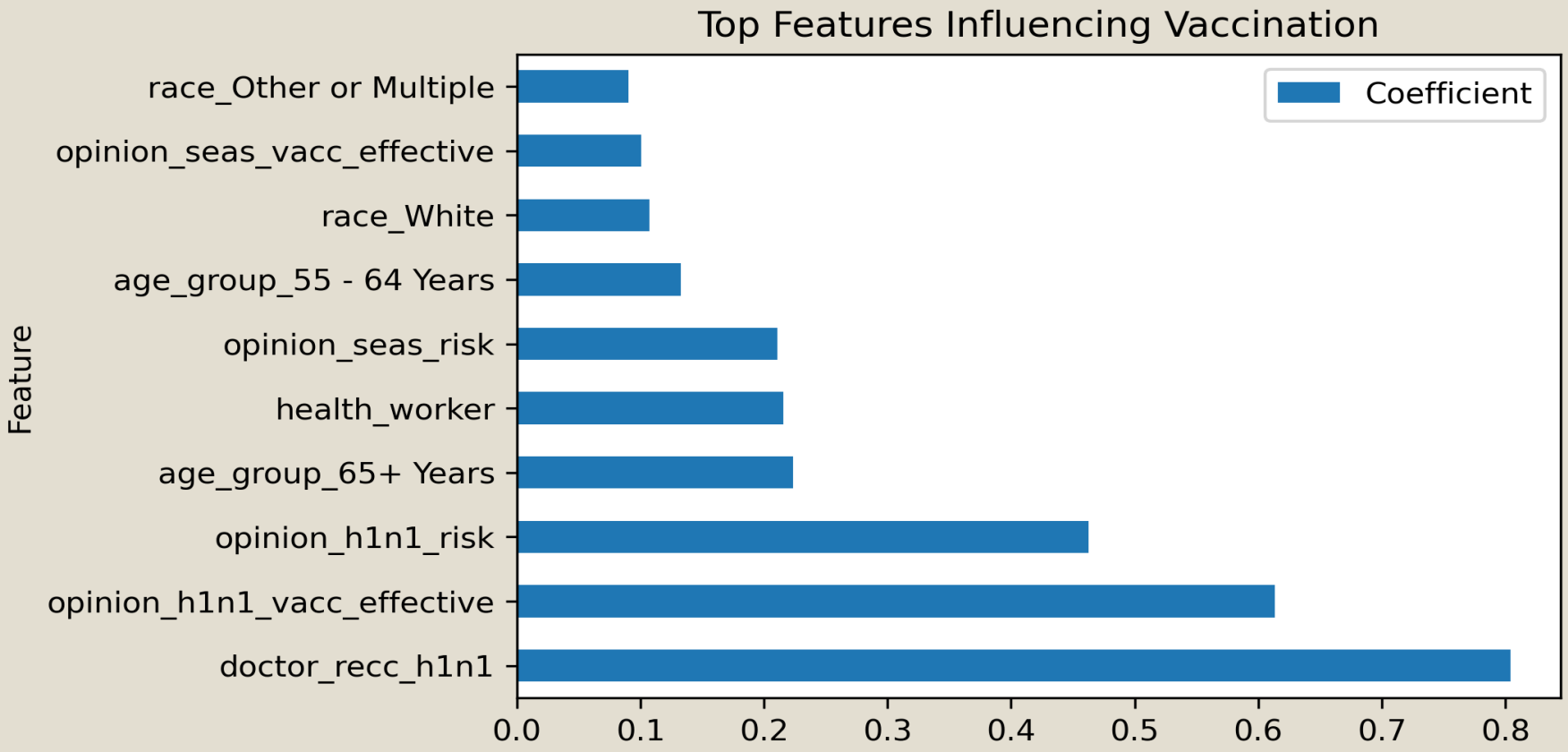
- Dataset Details:
  - Number of records: 26,707
  - Number of columns: 38
  - Target Variable: h1n1_vaccine (Vaccinated vs not Vaccinated)
- The dataset is imbalanced, with fewer vaccinated individuals.
- Dataset contains demographic and behavioral data



- The visualization displays how the data is imbalanced.

Key Influencing Features

- Both behavioral and perception-based factors strongly influence vaccination decisions.

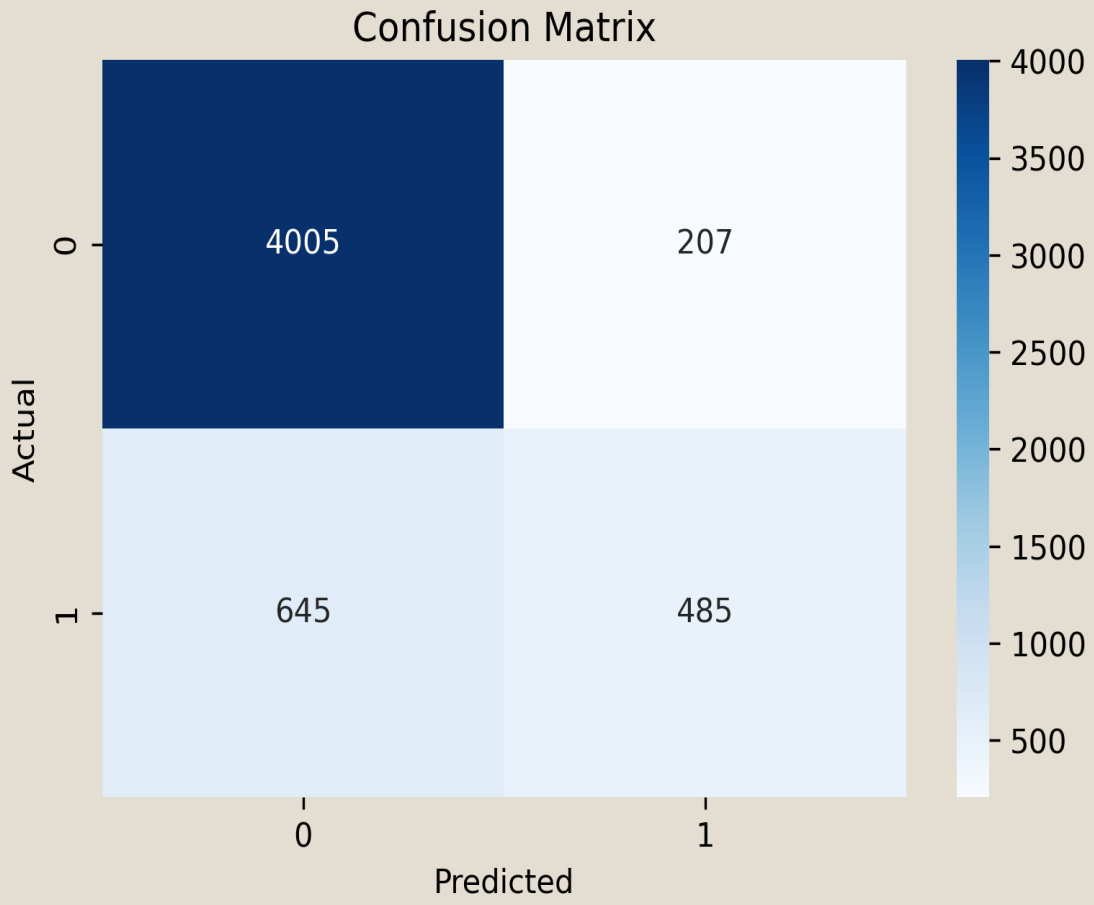


Modelling approach

- We used multiple models and improved them iteratively.
- The models used involve:

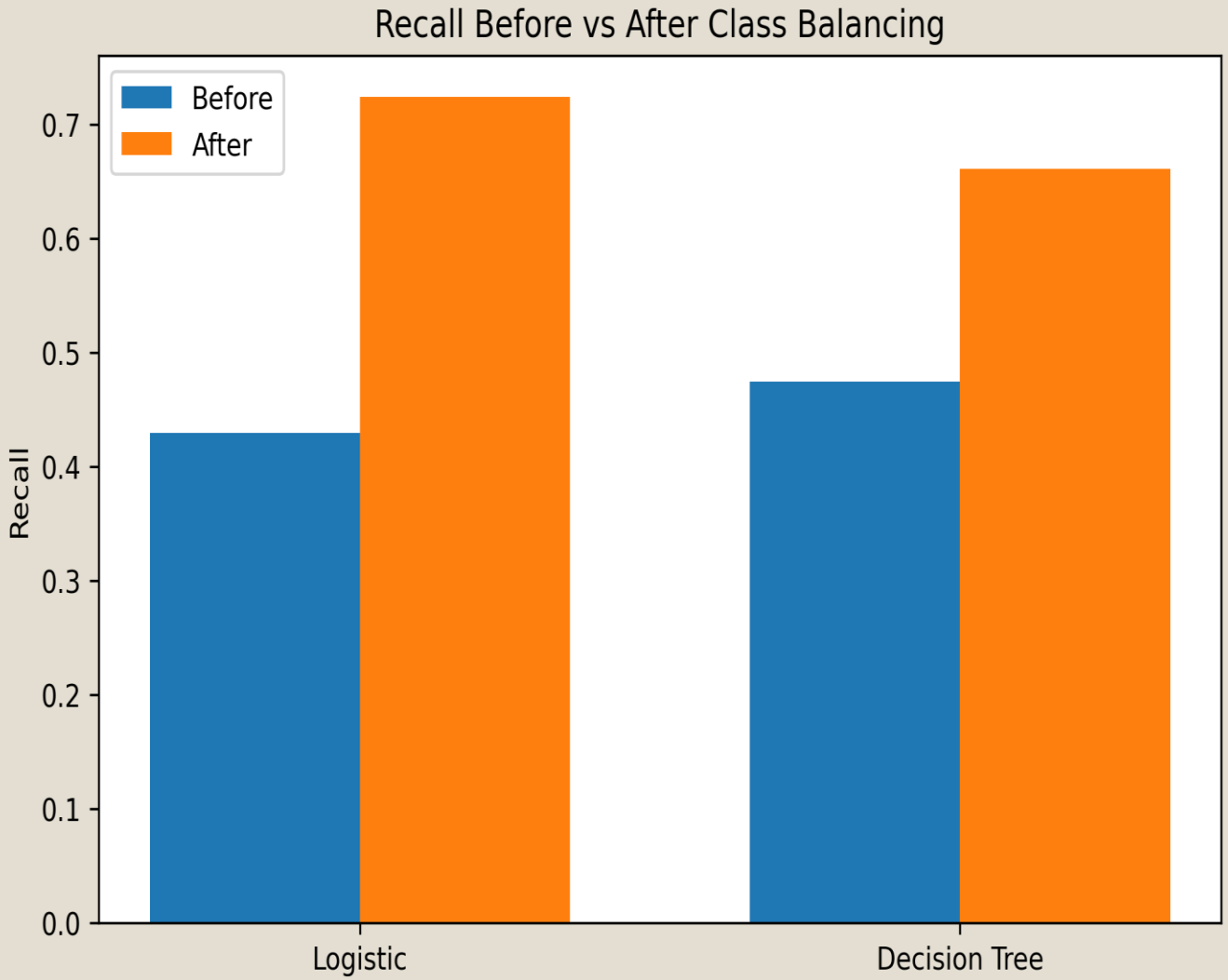
1. A logistic regression (Baseline Model)
2. A decision tree model
3. A balanced logistic regression
4. A balanced decision tree model

Baseline Model Performance



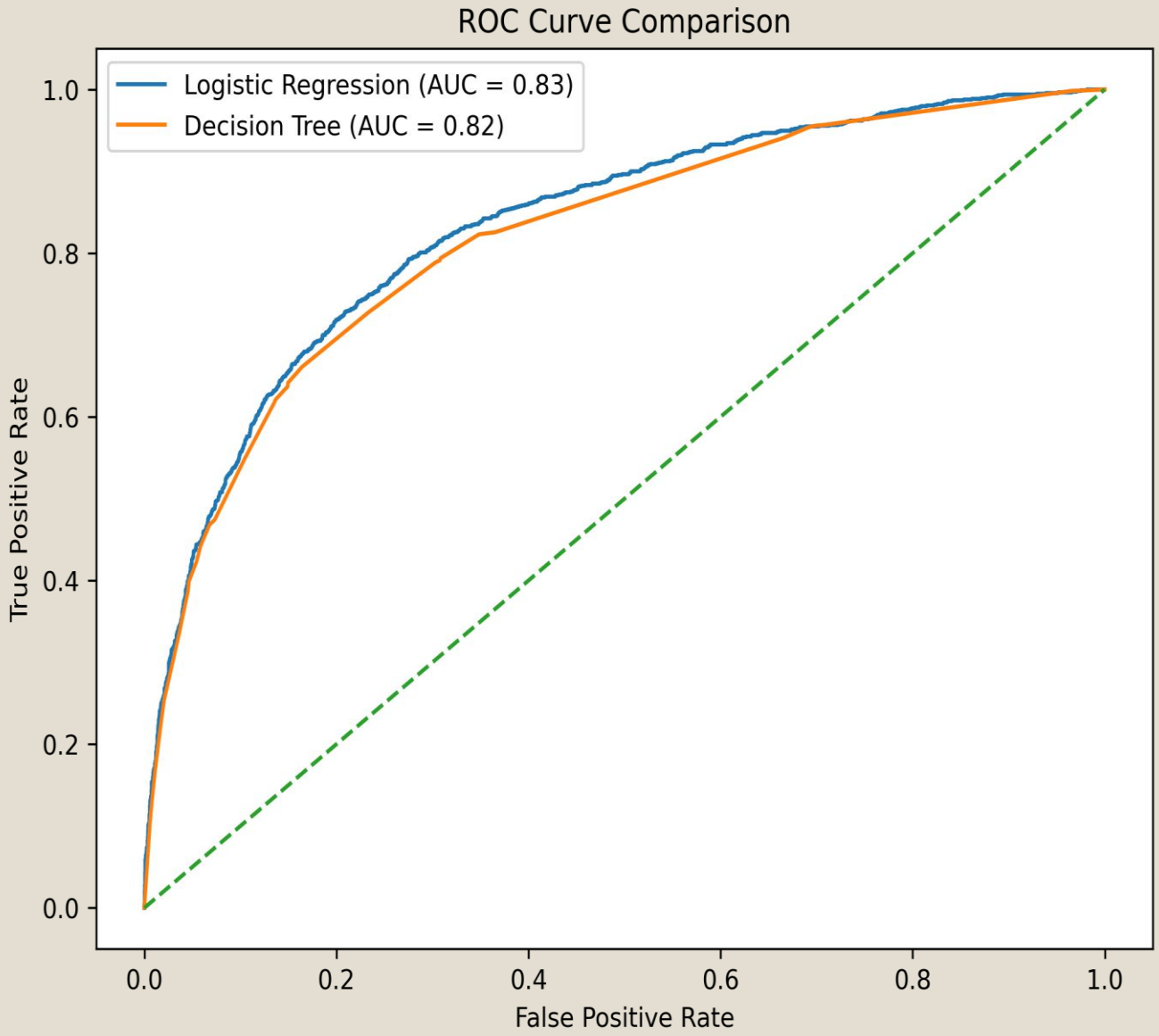
- The model performed well overall but struggled to identify vaccinated individuals.
- The model had an accuracy of 84%.

Improving the Model



- Due to the imbalance noticed in the data, class weighing was applied to dissipate the issue
- Handling class imbalance significantly improved performance of both models.
- The Logistic Regression showed greater improvement of 30%.

Model Comparison



- The Logistic Regression achieved a higher AUC (Area Under Curve) thus showing better overall classification ability.

Final Model Selection

Balanced Logistic Regression

- The model showed the highest improvement in recall (about 30% increase).
- The model is better at detecting the minority class.
- The model has a higher AUC.
- The model handles imbalance well.
- The model is more interpretable and best at identifying the vaccinated individuals.
- Thus this is the best model for predicting vaccination behavior and supporting targeted interventions.

Recommendations

- Increase doctor led vaccination campaigns.
- Improve public trust in vaccines.
- Target groups with lower vaccination likelihood.
- Use data driven strategies to improve outreach.
- Public health campaigns should focus on individuals less likely to get vaccinated.

Limitations

- Class imbalance affected the model performance.
- Data is self-reported and thus may contain bias.

THANK YOU!

- Any Questions?